LATIHAN TERSTRUKTUR (TUGAS KULIAH)

Pertemuan 4: Implementasi Unit Testing dengan GitHub Actions

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Kelas: Implementasi Pengujian P/L

Tanggal: 12 Oktober 2026

Repository: github.com/diva-28-ukay/latihan-git-1224026

BAGIAN A: GIT & GITHUB DASAR

Soal 1.1 Centralized vs Distributed: Centralized (CVCS): satu server pusat, client hanya working copy, commit butuh koneksi. Contoh: SVN, CVS. Distributed (DVCS): setiap clone full history, bisa commit offline lalu push/pull. Contoh: Git, Mercurial.

Soal 1.2 Lima Git commands: 1) git clone <url> - salin repo ke lokal. 2) git add <file> - masuk staging. 3) git commit -m pesan - simpan snapshot. 4) git push origin <branch> - kirim ke GitHub. 5) git pull - ambil+gabung remote.

Soal 1.3 .gitignore: daftar pola yang diabaikan Git. Mencegah target/, *.class, .idea/, .env masuk repo.

Soal 2 Praktik Git (sudah dijalankan, history: docs: add README, test: add test file, merge feature-test ke main):

```
git clone https://github.com/diva-28-ukay/latihan-git-1224026.git
cd latihan-git-1224026
git add README.md
git commit -m "docs: add README"
git push origin main
git branch feature-test
git checkout feature-test
git add test.txt
git commit -m "test: add test file"
git push origin feature-test
git checkout main
git merge feature-test
git push origin main
```

BAGIAN B: UNIT TESTING (Soal 3 & 4)

Soal 3 PendaftaranValidator.java: TC1/P1 nama kosong -> Nama harus diisi. TC2/P2 umur 16 -> Umur minimal 17 tahun. TC3/P3-P4 email null/tanpa @ -> Email tidak valid. TC4/P5 setuju=false -> Harus menyetujui syarat. TC5 sukses -> Pendaftaran berhasil. Test: PendaftaranValidatorTest.java. Run: mvn test -Dtest=PendaftaranValidatorTest

Soal 4 StudentGrade.java: 1) CC calculateGrade = 7 (predikat <0, >100, >=80, >=70, >=60, >=50 = 6 +1). 2) 7 paths: P1 <0 throw, P2 >100 throw, P3 >=80 A, P4 70-79 B, P5 60-69 C, P6 50-59 D, P7 <50 E. 3) Test StudentGradeTest.java (A/B/C/D/E + boundary + invalid + isPassed).

BAGIAN C: GITHUB ACTIONS (Soal 5 & 6)

Soal 5 Workflow Python CI: 1) Nama: Python CI. 2) Trigger: push main+develop, pull_request main. 3) strategy.matrix: job paralel python 3.10,3.11,3.12 via matrix.python-version. 4) Steps (hanya 4 di YAML): Checkout, Setup Python, Install dependencies, Run tests.

Soal 6 Workflow Java di .github/workflows/java-unit-test.yml:

```
name: Java Unit Test
on:
  push:
    branches: [main, develop]
  pull_request:
    branches: [main]
jobs:
  test:
    runs-on: ubuntu-latest
    steps:
      - name: Checkout code
        uses: actions/checkout@v4
      - name: Set up JDK 17
        uses: actions/setup-java@v4
        with:
          java-version: '17'
          distribution: 'temurin'
          cache: maven
      - name: Run tests
        run: mvn test
      - name: Generate coverage report
        run: mvn jacoco:report
      - name: Upload coverage report
        uses: actions/upload-artifact@v4
        with:
          name: jacoco-report
          path: target/site/jacoco/
```

BAGIAN D: COVERAGE (Soal 7)

TOTAL Stmts 50, Miss 11, 78%. 1) Total statement: 50. 2) Miss: 11. 3) Terendah: src/payment.py 60%. 4) Perbaiki: buka htmlcov, tambah test gagal/boundary/mock gateway, target >=80%. 5) 78% belum ideal (standar 80-90%), apalagi payment 60%. Layak baseline sementara, harus naik fokus ke payment.py.

BAGIAN E: BONUS BankAccount (Soal 8)

1) BankAccount.java: deposit, withdraw, getBalance. Validasi amount>0, withdraw<=saldo. 2) BankAccountTest.java 9 test: initial 0, initial 100, initial negatif throw, deposit 150, deposit negatif throw, withdraw 60, over-balance throw, withdraw negatif throw, transaksi 125. 3) Workflow: file yang sama (mvn test + jacoco + upload). 4) Repo: github.com/diva-28-ukay/latihan-git-1224026. Setelah push cek Actions hijau, download jacoco-report, screenshot coverage.

Run lokal: mvn test ; mvn jacoco:report ; buka target/site/jacoco/index.html

Push: cd latihan-git-1224026 ; git push origin main ; git push origin feature-test (pakai PAT jika diminta).